Token-Anchored State Editing in Language Models

Sahil Yadav

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Abstract

Where can the transient state of an in-context task be causally edited? Across Qwen, Mistral, DeepSeek-Llama, and Gemma, a residual value write applied where a fact is stated reproduces textual-counterfactual effects on retrieval, computation, and relational inference. We then perform an exhaustive mechanistic analysis in Qwen2.5-14B. In a controlled belief world, source-anchor writes match direct and inferred consequences, whereas transplanting complete matched state at a designated summary token moves at most 0.0003 of the natural effect despite 100% within-surface decoding. Locus controls show that the edited source anchor alone is sufficient through layer 32; its effect then falls as a query-specific readout becomes sufficient. On 30 held-out worlds in three longer prose styles, belief transfer replicates at 1.012 (95% CI [1.006, 1.018]), with the same checkpoint/readout dissociation. Prespecified behavioral gates fail for naturalized report and address-specificity arms, which we retain as a boundary. The evidence supports writable source anchors followed by a late query-conditioned handoff—a local causal mechanism, not proof that alternative or distributed state representations are absent.

1 Introduction

In-context learning makes a frozen transformer behave as if it had temporary variables. A prompt may bind a name to a value, introduce rules over that value, update an agent's belief, and query a downstream consequence. Behavioral success alone does not distinguish two computational pictures. The model might consolidate the relevant facts into a query-independent internal workspace, or it might repeatedly retrieve information from the tokens at which those facts were stated.

This distinction matters for interpretability and control. If transient state is consolidated, a single internal register may provide a natural target for editing. If source anchors remain causally privileged, reliable editing should instead be addressed: a value-like displacement must be written at the token that owns the value. Prior activation-steering work shows that residual directions can change behavior [16, 20, 23]; task-vector work shows that compact activations can mediate in-context functions [10, 19]. Neither fact by itself determines how a transformer stores bound, temporary data.

The distinction also has practical consequences. An editor aimed at the wrong site may change an answer without updating the state that supports related answers; a diagnostic may confuse readable traces with a component the model actually uses. Locating a write interface together with its downstream handoff therefore offers a stricter test for coherent editing, helps separate content failures from routing failures, and provides a way to evaluate whether a steering or safety intervention changes upstream task state or only the final output.

We introduce controlled tests that compare a textual counterfactual with an activation-space write. For a source value a and target value b, we extract neutral-carrier states $z(a)$ and $z(b)$ and add

$$\Delta_{a\to b} = z(b) - z(a) \tag{1}$$

at the prompt position where a occurs. The critical metric is not whether the output changes, but whether the intervention reproduces the effect of replacing a by b in the text, including downstream consequences and invariants.

Our evidence supports four increasingly strong observations:

1. **Writable value state.** Early residual differences behave as data writes across multiple computations and model families.
2. **Addressed propagation.** A write can alter a bound entity's direct and two-hop consequences while preserving unrelated state.

3. **Anchor/checkpoint dissociation.** In one behaviorally controlled belief world, the source anchor is writable while a designated query-independent checkpoint is causally inert under matched state transplantation.

4. **Query-conditioned handoff.** Minimal and multi-token locus interventions place sufficient causal support at source anchors through middle depth; that support then disappears as a late, query-specific readout becomes sufficient.

The last two observations form the paper's mechanistic hypothesis, not a universal architectural law. Our confirmatory protocol widens examples, tests late layers and query-specific sites, measures linear decodability with grouped held-out worlds, includes independent-family replications, and closes with a frozen held-out naturalized-surface test. The final test is deliberately reported as a split result: the central belief/checkpoint/readout contrast replicates, but two behavioral gates fail and therefore bound the broader claim.

The exhaustive probe, locus, and naturalized analyses are conducted on Qwen2.5-14B; Qwen2.5-7B, Mistral-7B, DeepSeek-Llama-8B, and Gemma-3-12B test which parts replicate across models. In one sentence, our main contribution is a depth-dependent causal account: prompt-bound values are editable at their source anchors through middle depth and become substitutable at a late, query-conditioned readout, while the tested summary checkpoint remains inert.

2 Related work

Activation intervention. Activation addition and representation engineering use contrastive residual directions to modulate model behavior [16, 20, 23]. Function and task vectors compress in-context demonstrations into reusable activations [10, 19]. Our object differs in role: the prompt supplies the program, while the intervention writes a value consumed by that program.

Localization and causal interpretation. Causal tracing and activation patching localize representations by replacing internal states between matched runs [9, 14, 22]. Localization depends on the corruption, metric, and intervention. Moreover, a successful low-dimensional intervention can recruit off-distribution or dormant mechanisms [12]. We therefore compare writes against natural counterfactual effects, require wrong-value and wrong-address controls, and do not equate decodability with causal storage.

Reliability and statistical variation. Steering effects can vary across inputs, layers, and evaluation criteria [2, 18], while causal interventions may create representations that diverge from the natural computation they are meant to model [7]. Mechanistic measurements should therefore be treated as statistical estimates over inputs, not properties certified by repeated intervention seeds [13]. Our frozen gates, distinct-row uncertainty, bidirectional swaps, and natural-textual upper bound operationalize these concerns. Recent demonstrations that single-position interventions can miss distributed output templates [3] specifically motivate our multi-token, full-prefix, and size-matched locus controls.

Editing versus locating knowledge. Knowledge-editing work has documented dissociations between the site at which information is localized and the site at which an edit is effective [8]. We study temporary, prompt-bound state rather than parametric factual knowledge, and use downstream computations to test whether the edited value is actually consumed.

Workspaces and binding. How transformers bind entities to attributes in context has been studied representationally through binding vectors [5] and, in concurrent work, through a compact attention circuit that writes entity identity at source positions and reinstates it at readout [15]; the latter is a mechanism-level account consistent with the causal picture we test here. Separately, recent interpretability work reports a reportable, causally swappable "global workspace" of active concepts during reasoning, while explicitly leaving open how those concepts are bound into relations [1]. Our checkpoint experiment addresses precisely that open question for prompt-bound state: we ask whether the *bound* configuration of a small world is causally substitutable at a query-independent position, and find that it is not, whereas the individual value anchors are. Finally, Shih et al. [17] show that models fine-tuned to write intermediate states causally reuse what they wrote; our verbalization experiment measures the complementary quantity in stock models—how a written statement redistributes causal load relative to the original anchors.

3 Experimental design

3.1 Identification strategy

For each row, a clean prompt contains source value a and a natural counterfactual changes only that occurrence to target b. Let $m(x) = \text{logit}(b) - \text{logit}(a)$ at the scored answer position. We report the natural effect

$$E_{\text{nat}} = m(x_b) - m(x_a), \tag{2}$$

the activation-write effect

$$E_{\text{write}} = m(x_a; \Delta_{a \to b}) - m(x_a), \tag{3}$$

and their ratio $\lambda = E_{\text{write}}/E_{\text{nat}}$. A ratio near one means effect matching, not merely output steering.

The identification assumption is local: neutral-carrier differences isolate a reusable value displacement at the chosen layer. We stress this assumption with same-norm random directions, embedding differences, wrong-value writes, and wrong-address writes. Behavioral gates require the model to solve both clean and natural prompts before an intervention is interpreted.

3.2 Task hierarchy

The workspace matrix contains retrieval, addition, subtraction, maximum, and a rule-defined label. Entity worlds separate value vocabulary from consequence vocabulary: changing Alice's city must change what she says through an in-context rule, while Bob's answer stays fixed. The structured-belief world contains four private beliefs and two true locations, followed by a literal `STATECHECK` marker before the question.

3.3 Anchor and checkpoint interventions

The anchor intervention adds $\Delta_{\text{Paris} \to \text{Rome}}$ at the token stating Alice's observed cube location. We measure her belief, what she tells a teammate, four invariant state variables, and a wrong-address edit at Bob's cube anchor.

For the checkpoint test, clean and natural runs are row matched. At each tested layer we transplant the complete residual state from the natural run into the clean run and vice versa. The primary site is `STATECHECK`. The final question token and answer-prefix readout are trajectory controls: activity there shows when query-specific answer state becomes causally transportable, but cannot turn the earlier marker into a query-independent buffer.

3.4 Confirmatory protocol

Discovery runs use 5–12 rows. The frozen confirmatory battery uses 30 distinct value transitions per workspace cell, 30 unique entity worlds, and all 30 structured-world nuisance pairs when tokenizer alignment permits. Workspace and entity generators use three genuine row seeds. The structured anchor uses one exhaustive row set; repeated random-direction seeds are not presented as independent data. Full details, gates, and stopping rules appear in the supplementary protocol. Model coverage comprises Qwen2.5-7B and Qwen2.5-14B [21], Mistral-7B [11], DeepSeek-R1-Distill-Llama-8B [4], and Gemma-3-12B [6]. The final closeout is fixed before execution: an exact edited-anchor-only sweep over 12 layers and 30 new Tokyo-to-Delhi worlds divided equally among case notes, witness transcripts, and curator narratives. No prompt, layer, surface, or model rescue is permitted after seeing this result.

Uncertainty is computed over distinct rows. We report paired row-bootstrap 95% intervals (10,000 resamples) for effect ratios and finite randomization p-values against prespecified same-norm directions. A causal result is interpreted only when clean and textual-counterfactual behavior pass the frozen accuracy gate; failure of that gate is an identification boundary, not a zero causal effect.

4 Results

4.1 A neutral value write is consumed by multiple computations

At layer 2, Qwen2.5-7B passes all 15/15 widened seed–cell tests, with effect ratios between 0.985 and 1.010. Mistral-7B likewise passes 15/15, with ratios between 0.939 and 1.004. Every passing cell also satisfies target-accuracy, wrong-value consequence, and same-norm random direction gates. The earlier Phi-3.5 pilot passes 4/4 compute cells but misses the frozen family-level criterion, illustrating why cell-level measurements and verdict logic are

Table 1: Confirmatory workspace results. Every seed–cell is evaluated on 30 distinct rows. Ratio ranges span behaviorally eligible tests; ineligible cells are excluded before causal adjudication.

Model	Seed–cells passed	Ratio range	Family verdict
Qwen2.5-7B	15/15	0.985–1.010	general (3/3 seeds)
Mistral-7B	15/15	0.939–1.004	general (3/3 seeds)
Phi-3.5-mini (pilot)	4/4 compute	—	below gate
DeepSeek-R1-Distill-Llama-8B	12/12 eligible	0.936–0.977	general (3/3 seeds)
Gemma-3-12B	15/15	0.982–1.000	general (3/3 seeds)

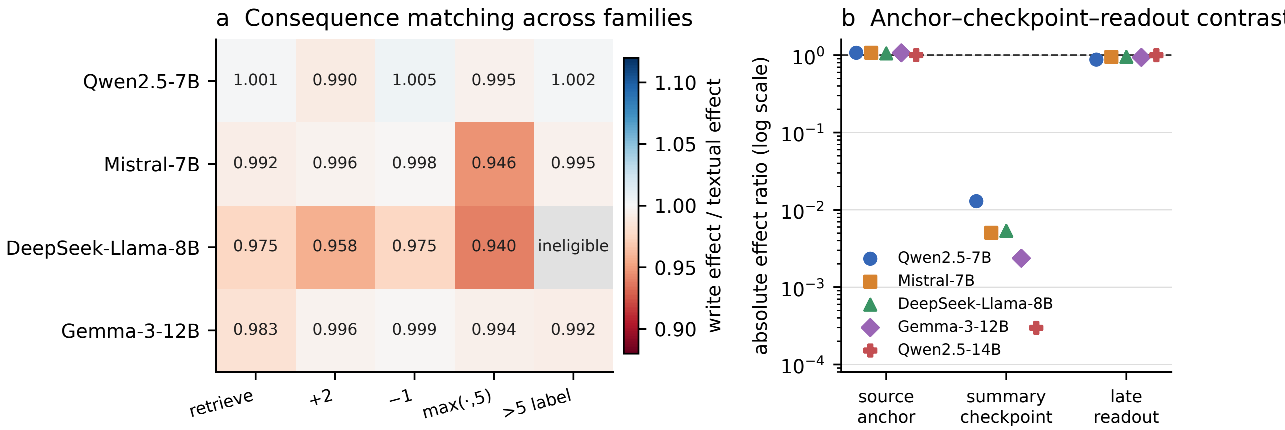

Figure 1: **Cross-family confirmation.** (a) Mean effect ratios for each workspace computation; gray denotes a failed behavioral gate rather than a causal negative. (b) Across five model/scale runs, source-anchor and late-readout effects are orders of magnitude larger than the maximum designated-checkpoint effect. The log scale makes the checkpoint values visible.

reported separately. DeepSeek-Llama passes all 12 eligible seed–cells; its rule-label cell is behaviorally ineligible across seeds. Gemma-3-12B provides the strongest independent-family replication, passing all 15/15 cells with ratios 0.982–1.000.

4.2 The write propagates through a two-hop world model

Across three widened Qwen2.5-7B city row sets, changing the representation of Alice's location changes the downstream word associated with that city. The intervention reproduces 0.974– 1.002 of the natural effect and reaches at least 96.7% target accuracy. The other-entity specificity cell passes in 2/3 seeds with exactly zero shift; the remaining seed is behaviorally ineligible, not a mechanistic failure. The keys surface and Mistral entity two-hop cells fail behavioral eligibility at $n = 30$ and are not counted as causal negatives. A Qwen2.5-14B pilot independently matches the city two-hop natural effect at 0.999. Gemma independently passes city retrieval and two-hop inference in all three row sets, with ratios 0.992– 1.007 and zero unrelated-entity shift. Its keys two-hop arm is behaviorally ineligible, reproducing the task rather than intervention boundary.

4.3 Source anchors and a designated checkpoint dissociate

In the structured belief world, the Qwen source-anchor write produces 1.075 of the natural belief effect and 1.052 of the reporting consequence over 30 worlds; the paired row-bootstrap 95% interval for the belief ratio is [1.068, 1.083]. The interval excludes 1: the write slightly but reliably *overshoots* the natural effect, which we attribute to the fixed neutral-carrier norm rather than a qualitative difference in mechanism; scaling calibration is straightforward future work, and the overshoot direction is conservative for our dissociation claim. Both consequences reach 100% target accuracy. Wrong-address editing remains specific and significant ($p = 0.01$). Three of four invariants remain perfect; the truth-sphere invariant falls to 70%, making Qwen's strict composite verdict partial.

Mistral supplies the clean independent-family confirmation on the mechanically aligned 18-world subset. The belief and reporting ratios are 1.076 (95% interval [1.062, 1.091]) and 1.038, all four invariants and wrong-address controls are perfect, and $p = 0.01$.

The exhaustive Qwen2.5-14B confirmation closes the earlier small pilot's strict composite verdict. Across 30 aligned worlds, the belief write reaches 1.001 of the natural effect (paired bootstrap 95% interval [1.000, 1.002]) and the reporting consequence reaches 0.993. Target accuracy, all four invariants, and address specificity are each 100%, yielding the frozen verdict `TOKEN_ANCHORED_CONFIRMED`.

Gemma-3-12B confirms the strict composite result in a second independent family. Its belief ratio is 1.073 (95% interval [1.066, 1.081]) and its reporting ratio is 1.013; all invariant and wrong-address gates remain at 100%. Its checkpoint maximum is only 0.0024, while its late readout reaches 0.936 with perfect forward and reverse accuracy at the final tested layer.

By contrast, full-state transplantation at `STATECHECK` reaches at most 0.0129 of the matched natural effect in Qwen and 0.0051 in Mistral over full-depth grids, and only 0.0003 across all 48 layers at Qwen2.5-14B. This is not a generic patching failure: at the query-specific final readout, the matched state becomes causally transportable late in depth, reaching 0.877, 0.950, and 0.999, respectively.

Nor is the checkpoint null explained by an absence of information. Nested grouped cross-validation decodes the eight held-out world labels at 100% accuracy on the ledger surface and 100% on a narrative paraphrase (chance 12.5%). Yet probes trained on one surface transfer to the other at only 16.7–18.8% at the checkpoint, while the anchor code transfers at 100% in both directions. Thus the checkpoint contains highly readable but surface-specific traces; decodability there does not imply a shared, causally substitutable relational register.

DeepSeek-R1-Distill-Llama-8B broadens the causal trajectory to a Llama-family architecture. Its intended anchor consequences are strong: belief ratio 1.055 (95% interval [1.034, 1.076]) and report ratio 1.013, both at 100% target accuracy. The strict anchor verdict is nevertheless partial: after the write, an unrelated same-valued belief has only 0% accuracy, and the wrong-address intervention fails its own-target gate. This is evidence for a portable content write but not a universally addressed binding interface. The checkpoint/readout dissociation survives this boundary: the full-depth checkpoint maximum is 0.0054, while the late readout reaches 0.953 with perfect forward and reverse accuracy.

4.4 Minimal causal support moves from the source to the readout

The checkpoint null leaves open a distributed-code objection: perhaps the correct query-independent state exists, but no single summary position is a sufficient intervention target. We therefore sweep matched-state swaps at the marker, the full local summary span, all six source anchors, cumulative and leave-one-out source sets, the entire pre-query prefix, and three seeded size-matched random loci. Exact full-prefix swaps reproduce the donor run at every layer, validating the intervention implementation.

From L2 through L32, the six source-anchor states are bidirectionally sufficient: forward ratios range from 0.999 to 1.012 and reverse ratios from 0.962 to 1.005, with 100% behavioral recovery. Removing the edited Alice-cube anchor collapses the effect to at most 0.031, whereas removing any other anchor preserves it. The marker, summary, and size-matched random maxima are respectively 0.0003, 0.0005, and 0.0032. The frozen exact single-anchor closeout then confirms that the edited anchor alone is sufficient over precisely the same L2–L32 interval. Its forward effect drops to 0.402 at L36, 0.245 at L41, and 0.001 at L46 while the query-specific readout rises. Thus the data identify a causal handoff rather than a permanently privileged token.

4.5 Held-out prose preserves the central contrast, with a boundary

The final frozen run changes both the transition (Tokyo to Delhi) and the surface form. Thirty new worlds are split evenly among longer case notes, witness transcripts, and curator narratives. The belief write matches 1.012 of the natural effect (95% interval [1.006, 1.018]), reaches 100% target accuracy, and exceeds all 20 same-norm random directions ($p = 0.048$). Across the same rows, the designated checkpoint moves at most 0.0021 of the natural effect and the late readout becomes sufficient from L41, reaching 1.001.

The composite closeout nevertheless returns `PAPER1_CLOSEOUT_BOUNDARY`. The model solves only 30% of the naturalized report counterfactuals, below the frozen 80% gate, despite a clean accuracy of 90% and a numerical write ratio of 0.996. The unrelated-belief control is likewise below gate before intervention (70% clean; 73% after the write). Consequently these rows cannot identify naturalized report transfer or address specificity. We retain the positive belief/checkpoint/readout result and report the two failed behavioral gates as the prespecified surface boundary; no prompt or model rescue is run.

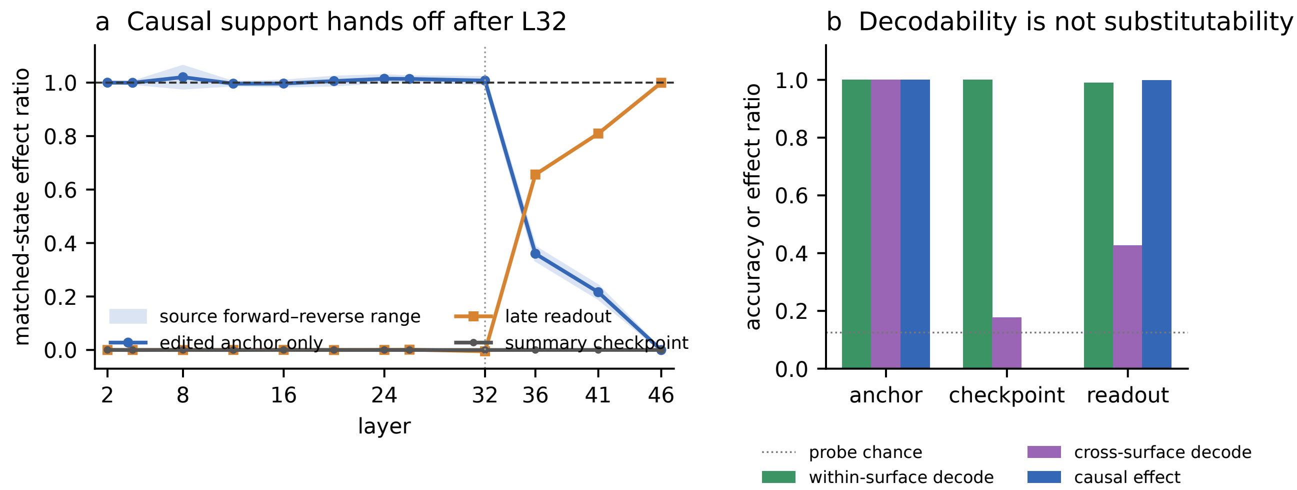

Figure 2: **A causal handoff and a decodability–substitutability dissociation.** (a) The exact edited source anchor is sufficient through L32 and then loses causal load as the late readout becomes sufficient; the designated checkpoint stays near zero. The band spans forward and reverse source swaps. (b) Within-surface linear decoding is perfect at both anchor and checkpoint, but cross-surface decoding and causal substitution sharply distinguish them.

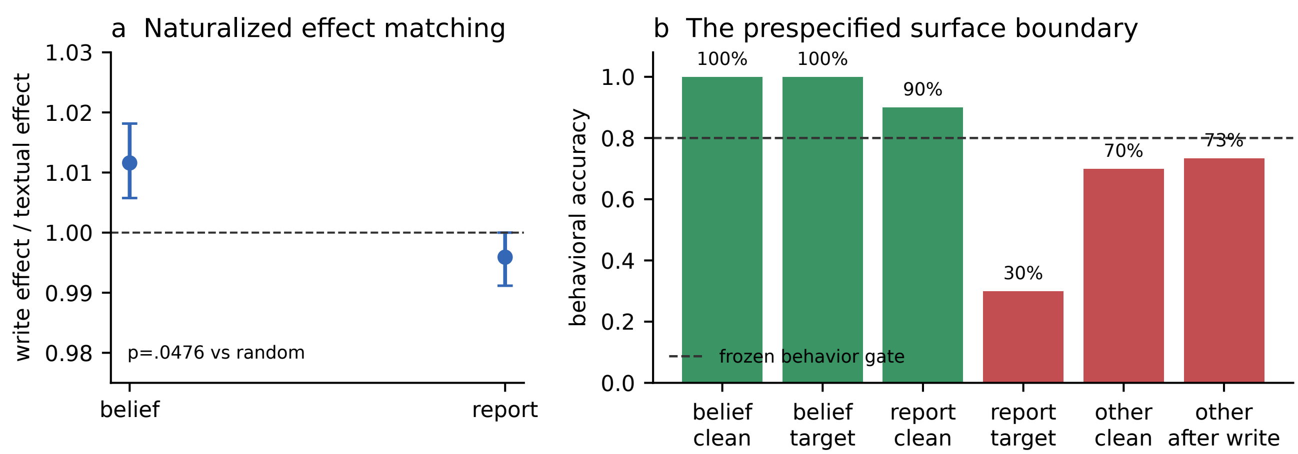

Figure 3: **Held-out naturalized closeout.** (a) Paired-bootstrap 95% intervals for the two numerical effect ratios. Belief transfer is behaviorally eligible and beats the random-direction panel. The report ratio is shown descriptively only. (b) Frozen behavioral gates explain the split verdict: belief passes, while the report target and unrelated-belief baseline do not. Red bars are identification failures, not causal zeros.

4.6 A reverse-base control rejects the quorum interpretation

Without an explicit verbalized belief, the history-anchor write retains 1.004 of the natural effect. Adding a contradictory verbalization reduces the history-only intervention to 0.217, while editing both history and verbalization restores 0.997. This initially admitted a corroborating-witness or quorum interpretation, but that interpretation was confounded with a Paris/default prior. The pre-registered reverse-base control begins from two Rome-consistent sites and applies the Paris-minus-Rome write. Editing history alone produces 0.669 of the full natural effect, editing verbalization alone produces 0.781, and editing both produces 1.000; both one-site conditions select Paris rather than the untouched Rome witness. The frozen verdict is `PARIS_PRIOR_REPLICATES_REVERSE_BASE`. We therefore reject the quorum and tamper-evidence claims. Verbalization changes the relative efficacy of the two sites, but this asymmetric redistribution remains a boundary result rather than evidence for a cleanly relocated memory register.

5 What the experiments do and do not establish

The results support an operational statement: for the tested tasks, source tokens expose a writable value interface whose effects propagate through downstream computation. Address specificity holds in Qwen, Mistral, and Gemma structured worlds but fails in the DeepSeek-Llama run, so content efficacy and routing specificity must be separated. The checkpoint intervention is inert across four model families even though query-specific state becomes causally sufficient at the late readout. At 14B, the state is linearly readable at the summary position without that position serving as a causally sufficient register under matched state transplantation. Poor cross-surface transfer further shows that this readable checkpoint code is not a common surface-invariant coordinate system.

The multi-token controls sharpen “token anchored” into a depth-dependent claim. Early and middle layers admit a minimal sufficient intervention at the edited source anchor; the summary span and arbitrary same-sized positions do not. After L32 the anchor loses sufficiency while the readout gains it. This supports repeated access followed by query-conditioned transport, but does not show that the anchor is the only necessary state at every step or identify the attention path that performs the transport.

This does not imply that state is literally stored in one token, that the model uses no distributed representation, or that every internal edit follows the model’s natural computation. Autoregressive attention can repeatedly access the prompt; source-token interventions may alter many later activations. The DeepSeek-Llama spillover shows directly that content efficacy does not guarantee binding specificity. Our claim concerns causal access and substitutability under specified interventions. The naturalized closeout adds a second boundary: surface changes preserve the belief/checkpoint/readout contrast, but they can also make a downstream report or invariant control behaviorally unidentifiable. Broader surface robustness therefore cannot be inferred from the positive belief arm.

6 Limitations

The tasks use synthetic, controlled worlds and single-token values. The three naturalized prose styles vary wording and length but are not a natural-corpus benchmark. Models are open-weight, at most 14B parameters, and mostly instruction tuned. Several runs use 8-bit, NF4, or AWQ weights; residual interventions remain in floating point, but an unquantized replication is not available at the largest scale. A failed patch is conditional on site, layer, position, and intervention semantics; it is not proof of representational absence. Linear probes establish decodability, not use, and the cross-surface probe result is measured only on Qwen2.5-14B. The locus experiment establishes sufficiency and one leave-one-out necessity result, not a complete minimal causal graph. Finally, pilots were selected during an exploratory sequence. We keep them separate from frozen confirmation and release row-level artifacts, configs, exclusions, and hashes so that this distinction is auditable.

7 Ethics and broader impacts

The experiments use synthetic prompts, open-weight models, and no personal or human-subject data. More reliable activation editing could improve debugging and causal auditing, but it could also support covert steering or create false confidence when a low-dimensional intervention succeeds off distribution. We therefore release negative controls and failed eligibility gates alongside positive results, and frame the method as a local diagnostic rather than a general-purpose truth or safety editor.

8 Conclusion

Controlled residual writes reveal where a bound value can be changed so that its consequences update coherently. Across Qwen, Mistral, DeepSeek-Llama, and Gemma, source anchors dominate a designated query-independent summary checkpoint. At 14B, exact and multi-token controls identify the stronger sequence: a single edited anchor is causally sufficient through middle depth, then loses sufficiency as a late query-specific readout forms. Perfect within-surface decoding at the inert checkpoint makes the result a direct dissociation between readable information and causal substitutability. The DeepSeek address spillover and naturalized behavioral failures prevent a universal storage claim. Together the results support token-anchored access followed by query-conditioned transport as a concrete mechanism for temporary state in the tested tasks, while leaving distributed representations and the routing circuit itself open.

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A Frozen gates and artifact policy

The full pre-registration is distributed with the source as `PREPRINT_CONFIRMATORY_PROTOCOL.md`. Every output stores generated rows, model identifier, quantization, layer grid, gates, per-row effects, and random-control summaries. Failed behavioral gates are labeled ineligible. Discovery pilots and confirmatory artifacts occupy separate directories and are not pooled.

A.1 Behavioral and causal gates

Unless a protocol specifies a stricter condition, clean and natural textual counterfactual accuracy must each reach 80% before the causal cell is interpreted. A sufficient bidirectional matched-state swap requires a forward effect ratio in $[0.7, 1.3]$, a reverse ratio in the same interval, and at least 80% donor-target and clean-state recovery. Checkpoint inertness requires the maximum absolute ratio to stay below 0.3. Random-direction tests use prespecified, norm-matched draws and the finite-sample value $(1+\#\{T_0 \geq T_{\text{obs}}\})/(1+N_0)$. No ineligible cell contributes to a pass/fail causal count.

A.2 Models and mechanism coverage

Model	Size/runtime	Workspace	Anchor/checkpoint
Qwen2.5-Instruct	7B, 8-bit	yes	yes
Mistral-Instruct v0.3	7B, 8-bit	yes	yes
DeepSeek-R1-Distill-Llama	8B, quantized	yes	yes
Gemma-3-Instruct	12B, quantized	yes	yes
Qwen2.5-Instruct-AWQ	14B, AWQ	—	exhaustive

Table 2: Frozen confirmatory model coverage. The 14B run concentrates compute on the exhaustive mechanism, locus, probe, and closeout arms rather than repeating the widened workspace matrix.

The final Qwen2.5-14B kernels ran on two Tesla T4 GPUs. The archived provenance record identifies Python 3.12.13, PyTorch 2.10.0+cu128, Transformers 5.0.0, Accelerate 1.13.0, and GPTQModel 7.1.0, together with the exact Kaggle model source. Each artifact additionally stores model key, quantization recipe, layer grid, row count, random seed, and protocol version.

A.3 Evidence regeneration and integrity

The confirmatory ledger contains seven immutable JSON artifacts: Qwen-7B, Mistral-7B, Qwen-14B, DeepSeek-Llama-8B, Gemma-3-12B, the Qwen-14B locus sweep, and the final closeout. A deterministic audit verifies file hashes, row counts and uniqueness, tokenizer-selected exclusions, layer coverage, full-prefix sanity controls, null panels, and the final adjudicated verdict. The final manifest status is `PAPER1_EVIDENCE_FROZEN_WITH_BOUNDARY`. The archived closeout artifact has SHA-256 `597F79E4C253A5ECF7F8440EDD061ABFD5155FACC221C7CFD66DEA445FB59422`. Running `rebuild_evidence.py` recomputes all paired-bootstrap summaries, regenerates the TeX macros used in this paper, and reruns the audit before returning success. `build_figures.py` reads only those immutable artifacts and the generated uncertainty summary.

A.4 Closeout surface construction

The naturalized closeout uses 30 held-out nuisance worlds for the unseen Tokyo-to-Delhi transition. Ten rows each are rendered as a case note, witness transcript, or curator narrative. Clean and natural prompts differ only at the aligned source value. Tokenizer preflight checks single source anchors, stable query positions, and absence of target leakage before weights are loaded. The report and unrelated-belief arms are retained in the artifact despite failing behavioral gates; no replacement rows or alternate phrasings are introduced.